



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Pathology of Hemorrhage

Principles and Practices of Osteopathic Medicine I

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Heal.
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Learning Objectives – Pathology of Hemorrhage

1. After a brief background description of normal hemostasis, discuss and describe disorders of primary hemostasis beginning with **quantitative platelet deficiencies including thrombocytopenia and immune thrombocytopenia purpura (acute and chronic)**
2. Discuss, describe, and compare the pathology of quantitative platelet deficiencies which include **thrombotic thrombocytopenic purpura and hemolytic uremic syndrome.**
3. Discuss, describe and compare **the qualitative platelet dysfunctional disorders Bernard-Soulier disease and Glanzmann thrombasthenia**
4. Discuss, describe, and compare the pathology of qualitative platelet dysfunctional disorders which include disorders of platelet secretion, aspirin, and uremia.
5. Discuss, discuss, and compare bleeding disorders caused by clotting factor deficiencies which include **von Willebrand disease, Hemophilia A, Hemophilia B, and Vitamin K deficiency.**
6. Describe and discuss the pathology (includes pathogenesis, etiologies, morphology, clinical and lab findings) of **disseminated intravascular coagulation**, disorders of fibrinolysis, and bleeding caused by vessel wall abnormalities.

Pathology of Hemorrhage

Overview of Bleeding Disorders

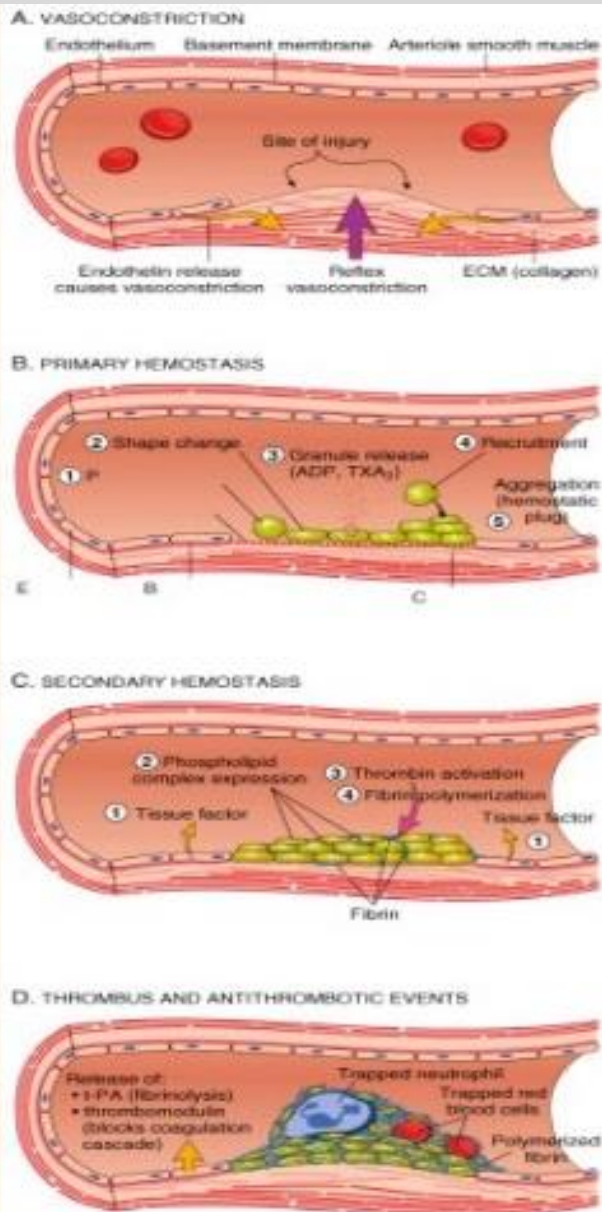
- **Platelet Deficiency (Quantitative)**
 - **Thrombocytopenia**
 - **Immune thrombocytopenia purpura (ITP)**
 - **Thrombotic thrombocytopenic purpura (TTP)**
 - **Hemolytic Uremic Syndrome**
- **Platelet Dysfunction (Qualitative)**
 - **Bernard Soulier Disease**
 - **Glanzmann's Thrombasthenia**
 - Disorders of Platelet secretion
 - Aspirin
 - Uremia
- **Clotting Factor Deficiency / Dysfunction**
 - **Von Willenbrand's Disease**
 - **Hemophilia A / B / C**
 - **Vitamin K Deficiency**
 - **Disseminated Intravascular Coagulation**
- **Disorders of Fibrinolysis**
 - Plasmin over activity
 - Reduced α 2-antiplasmin
- **Vessel wall abnormalities**
 - Infectious
 - Drug reaction
 - Scurvy (Vit C def) / Ehlers-Danlos / Cushings
 - Henoch-Schonlein Purpura
 - Hereditary Hemorrhagic Telangiectasia
 - Perivascular Amyloidosis

Hemostasis

1. Arteriolar Vasoconstriction
2. Primary Hemostasis (Platelet Plug) -
adhesion, shape change, secretion of granules
to induce aggregation, recruitment,
aggregation
3. Secondary Hemostasis (Coagulation Cascade)
4. Clot Stabilization and Anti-Thrombotic Events

Following a vascular injury, platelets adhere and aggregate to form the **primary** hemostatic plug and also promote key reactions in the coagulation cascade that lead to **secondary** hemostasis and formation of a fibrin clot

SUMMARY



Diagrammatic representation of the normal hemostatic process:

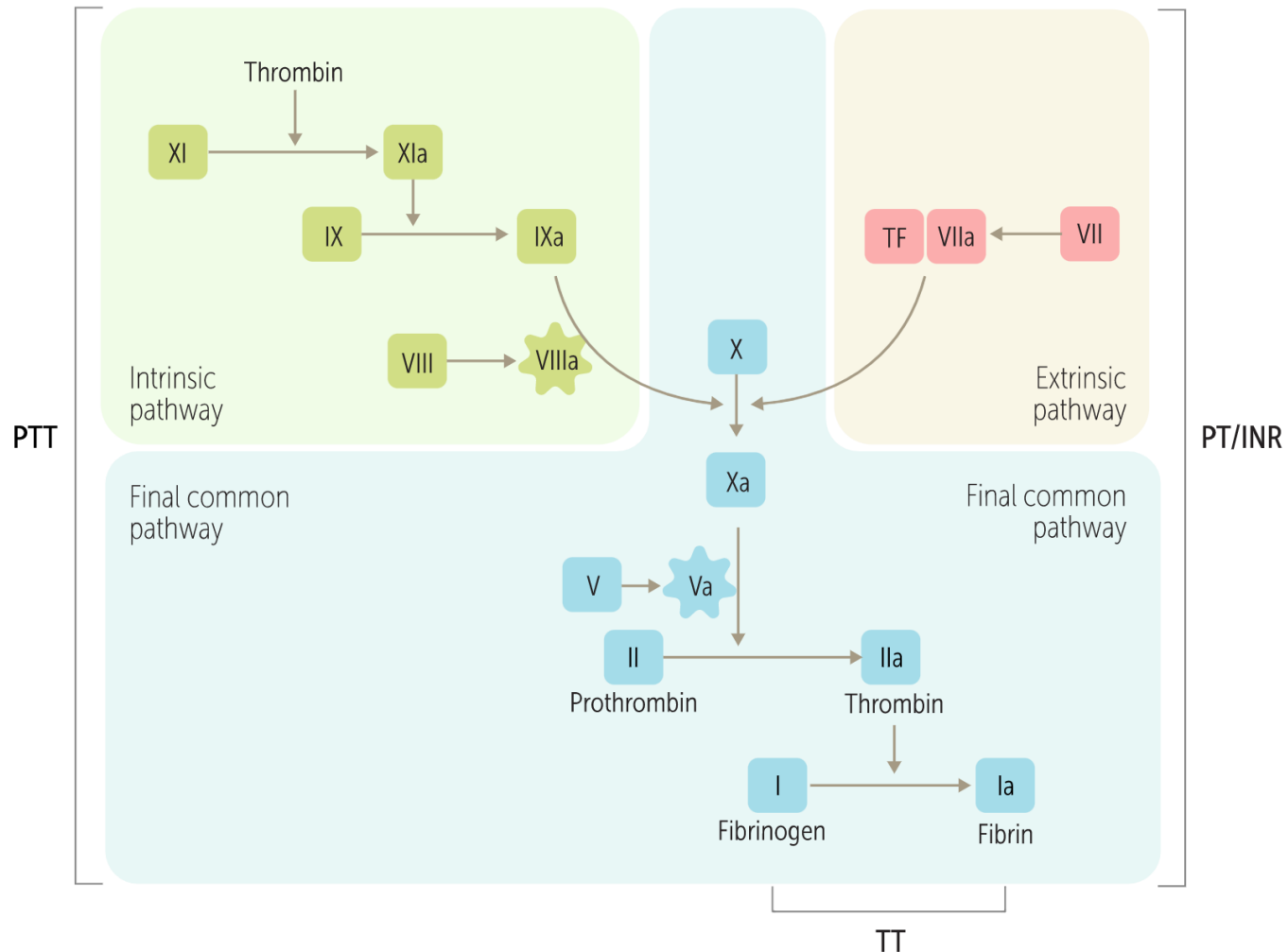
A, transient **vasoconstriction**:

B, Platelets adhere to exposed extracellular matrix (ECM) via von Willebrand factor (vWF) and are activated, undergoing a shape change and granule release; released adenosine diphosphate (ADP) and thromboxane A₂ (TxA₂) lead to further platelet aggregation to form the **primary hemostatic plug**.

C, Local activation of the coagulation cascade (involving tissue factor and platelet phospholipids) results in fibrin polymerization, "cementing" the platelets into a definitive **secondary hemostatic plug**.

D, Counter-regulatory mechanisms, such as release of tissue type plasminogen activator (t-PA) (fibrinolytic) and thrombomodulin (interfering with the coagulation cascade), **limit the hemostatic process to the site of injury**.

Physiology of Secondary Hemostasis Coagulation Cascade (REVIEW)

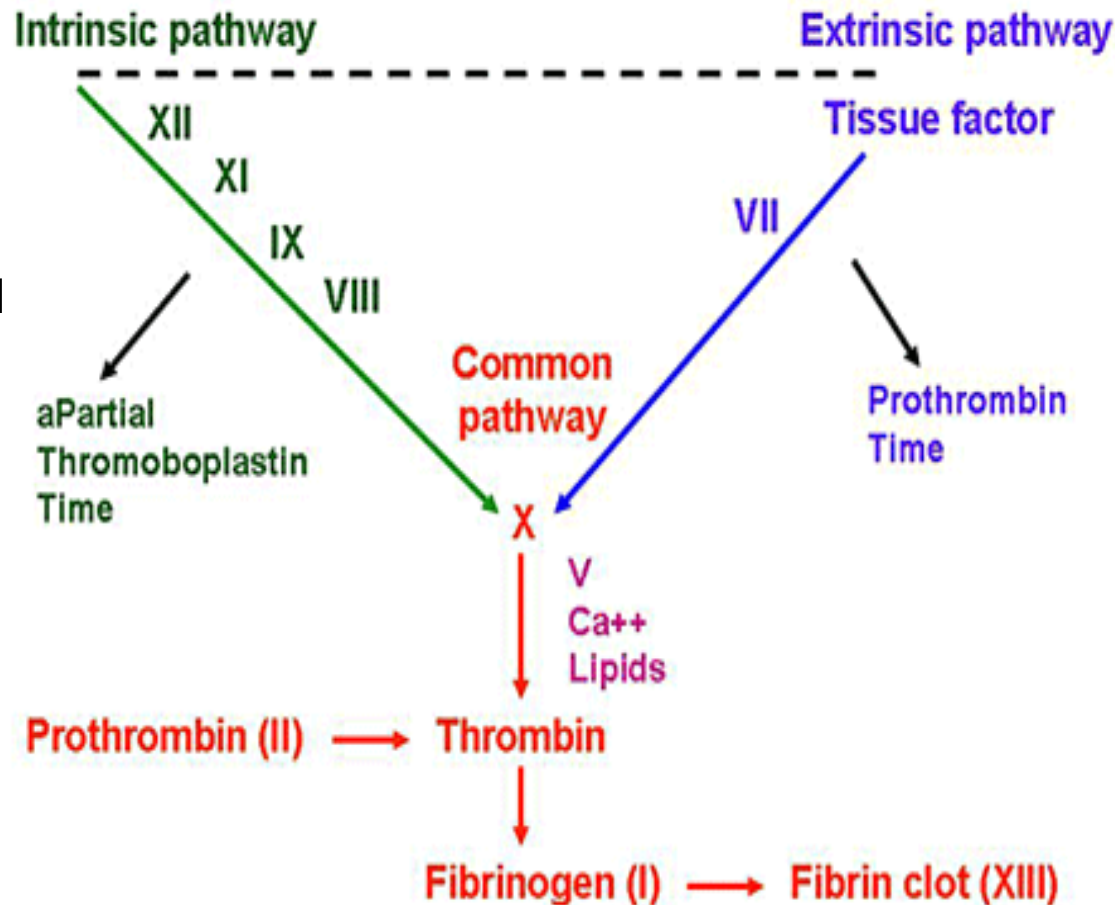


Intrinsic pathway:
activated by
damage inside
vasculature.

Platelets,
exposed/damaged
endothelium, sube
ndothelial
collagen, and
other chemicals
can activate this
arm of pathway;

**Slower than the
extrinsic pathway,**
thought to
generate a more
robust
coagulation
response.

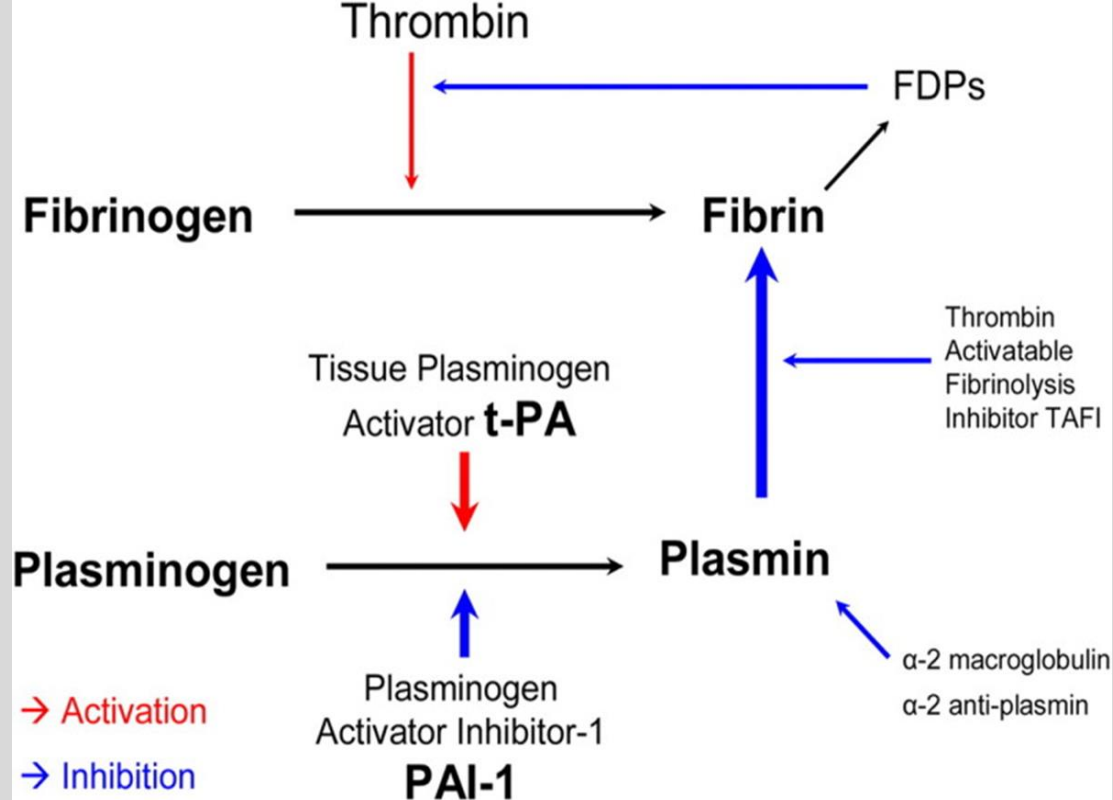
Coagulation Cascade



Extrinsic pathway:
activated by
trauma that
causes blood
to leave
vasculature.

**Tissue
thromblastin
factor is
activated and
sets off
cascade**

**Quicker than
the intrinsic
pathway**



LIMIT THE HEMOSTATIC PROCESS TO THE SITE OF THE INJURY

- **Tissue plasminogen activator converts plasminogen to plasmin**
- **Plasmin cleaves fibrin and serum fibrinogen, destroys coagulation factors, and blocks platelet aggregation.**
- **Alpha-2-antiplasmin inactivates plasmin**

Coagulation Disorders: the coagulation panel - PT, PTT, INR (REVIEW)

Prothrombin time (PT): time to clot (seconds), mainly measure of extrinsic pathway (factor VII)

International normalizing ratio (INR): used to normalize PT and compare INR between different institutions (which have different labs)

Calculation based on results of PT, used to monitor patients being treated with warfarin (anticoagulant)

Activated partial thromboplastin time (aPTT or just PTT): time to clot (seconds), measure of intrinsic pathway

Thrombin time (TT): time to clot, dependent on thrombin catalyzing fibrin production - not really used

Bleeding time (BT): time it takes for a patient to stop bleeding, looks for platelet dysfunction - not really used; all platelet disorders have increased bleeding time

Coagulation Disorders: coagulation panel – Fibrinogen & D-Dimer (REVIEW)

- **Fibrinogen:** glycoprotein synthesized by the liver, it is converted to fibrin (clot structure) by thrombin
- **D-dimer:** major product of fibrinolysis (fibrin degradation products, FDPs), elevated serum levels indicate recent intravascular coagulation
 - Not very specific

Bleeding Disorders: “Hemorrhagic Diatheses”

Excessive bleeding:

(1) increased fragility of vessels

(2) platelet deficiency or dysfunction

**(3) derangement of coagulation,
alone or in combination**

Quantitative **Platelet** Disorders

Thrombocytopenia

Immune thrombocytopenic purpura
(chronic and acute)

Thrombotic microangiopathies:
Thrombotic Thrombocytopenic Purpura
Hemolytic Uremic Syndrome
(typical and atypical)

Skin Signs of Bleeding

Petechiae: red/purple dots that represent bleeding from **leaking capillaries**

- Smallest in size ($<1\text{mm}$)
- Usually sign of primary hemostasis problem



Skin Signs of Bleeding

Purpura: petechiae which have coalesced and become bigger

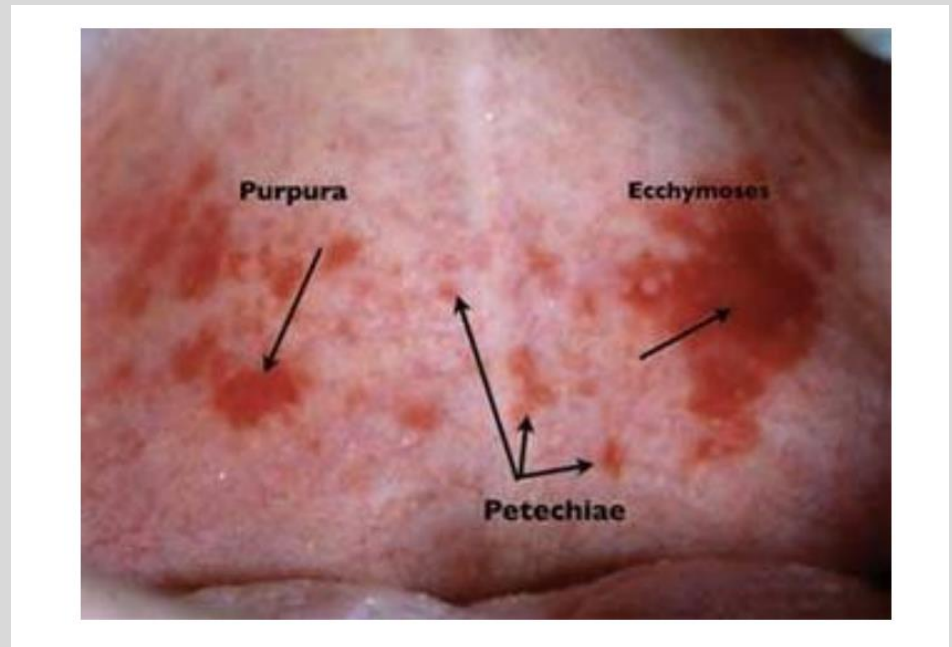
- “Medium” size ($>1\text{mm}$ & $<10\text{mm}$)
- Palpable versus non-palpable



Skin Signs of Bleeding

Ecchymosis: occurs deeper in the dermal layer

- “Large” in size ($>10\text{mm}/1\text{cm}$)
- Usually sign of secondary hemostasis problem



THROMBOCYTOPENIA

Thrombocytopenia < 150,000 /microliter (N = 150,000-450,000/ul)

- Platelets < 50,000 – Minor Trauma = Bleed; increased risk of bleeding during surgery
- Platelets < **20,000** – Spontaneous Bleeding; intracranial bleeding possible
- **Normal PT/PTT**
- **Petechiae**: common sign of **quantitative problems**, not commonly seen in qualitative
- Present with mucous membrane bleeding and microhemorrhages (petechiae, epistaxis (nose bleeding)), heavy menses, **oral/mucosal bleeding (petechiae, purpura)**
- Check platelet count (decreased), bleeding time (increased), smear, marrow

Causes of Thrombocytopenia (decreased platelet count)

Decreased Production of Platelets

- Selective impairment of platelet production
- Drug-induced: alcohol, thiazides, cytotoxic drugs
- Infections: measles, human immunodeficiency virus (HIV)
- Nutritional deficiencies: B¹², folate deficiency (megaloblastic leukemia)

- Aplastic anemia
- Bone marrow replacement
- Leukemia, disseminated cancer, granulomatous disease

- Ineffective hematopoiesis
- Myelodysplastic syndromes

Decreased Platelet Survival

- Immunologic destruction
 - **Chronic immune thrombocytopenic purpura**
 - **Acute immune thrombocytopenic purpura**
 - Systemic lupus erythematosus, B-cell lymphoid neoplasms
 - Alloimmune: posttransfusion and neonatal
 - Drug-associated: quinidine, heparin, sulfa compounds
 - Infections: HIV, infectious mononucleosis (transient, mild), dengue fever
- Nonimmunologic destruction
 - **Disseminated intravascular coagulation**
 - **Thrombotic microangiopathies: TTP, HUS**
 - Giant hemangiomas

Sequestration

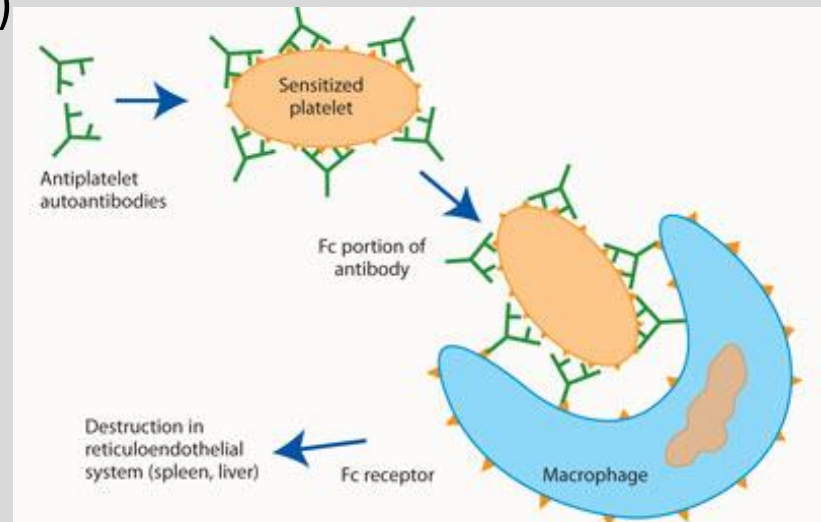
- Hypersplenism

Dilution

- Transfusions

Chronic Immune Thrombocytopenic Purpura

- Platelet Deficiency (**Quantitative**) – 1° Hemostasis
- Caused by **autoantibody-mediated destruction of platelets**, primary (idiopathic) or secondary; **diagnosis of exclusion** (R/O secondary causes)
- Pathogenesis: Autoantibodies usually directed against **platelet membrane glycoproteins IIb-IIIa or Ib-IX present in plasma and bound to platelet surface** in about 80% of patients
- Antiplatelet antibodies act as opsonins recognized by IgG Fc receptors expressed on phagocytes; leads to **increased platelet destruction**
- **Thrombocytopenia usually greatly improved by splenectomy** (spleen major site of removal of opsonized platelets; possible removal of a source of autoantibodies since splenic red pulp has many plasma cells)



Robbins & Cotran Pathologic Basis of Disease, 10th ed., 2021, Ch. 14

CHRONIC ITP - Platelet Deficiency (Quantitative) – 1° Hemostasis

- Primary chronic ITP: idiopathic/immune condition
- Secondary: SLE, HIV, CLL
- Onset: insidious
 - bleeding into skin and mucosal surfaces
 - pinpoint hemorrhages (petechiae), esp. in dependent areas (capillary pressure higher)
 - petechiae may become confluent, resulting in purpura
 - history of easy bruising, nosebleeds (epistaxis) and gingival bleeding
 - primary manifestation may be: melena, hematuria, or excessive menstrual flow

CHRONIC ITP

- Spleen usually normal size; congestion of sinusoids and enlargement of splenic follicles
- Bone marrow: modestly increased number of megakaryocytes, some immature, with large, nonlobulated, single nuclei (not specific finding, reflects accelerated thrombopoiesis because of platelet destruction)
- **Skin and mucous membranes: petechial bleeds**
- **Peripheral blood: abnormally large platelets** (megathrombocytes) (sign of accelerated thrombopoiesis)

Chronic ITP

- Labs: **low platelet count**, normal or increased megakaryocytes in bone marrow, **large platelets in peripheral blood** (evidence of accelerated platelet destruction); **PT and PTT normal**
- Tests for platelet autoantibodies have low sensitivity and specificity, not clinically useful; **diagnosis of exclusion**
- Most patients respond to **glucocorticoids** (inhibit phagocyte function), but may relapse after withdrawal
- **Splenectomy** for severe thrombocytopenia normalizes platelet count in about two-thirds of patients, but **increases risk of bacterial sepsis**

Acute Immune Thrombocytopenic Purpura

- Also caused by **autoantibodies to platelets**, but clinical features and course distinct
- Mainly **disease of childhood** , esp. 2 – 5 years old
 - **Most common cause of isolated thrombocytopenia in pediatric patients**
- Symptoms appear abruptly, **often 1 to 2 weeks after self-limited viral illness** (may trigger development of autoantibodies via uncertain mechanisms)
- **Self-limited (v. chronic ITP)**, usually resolves spontaneously within 6 months
- Glucocorticoids given only for severe thrombocytopenia
- In about 20% of children (usually those without viral prodrome) thrombocytopenia persists; childhood form of chronic ITP, follow course similar to adults

Acute Immune Thrombotic Thrombocytopenia (ITP) - Pathogenesis

- *Pediatric patient (typically)*
- Sudden onset widely spread **petechiae/bruising** over the entire **body**, which is not gravity-dependent
- Platelets usually $< 20,000/\mu\text{L}$
- Signs and symptoms of mucocutaneous bleeding (petechiae, bruising, mucosal bleeding, especially oral, and/or epistaxis)
- **IgG autoantibodies** formed against platelet surface proteins causes destruction of platelets in the **spleen and liver reticuloendothelial system**
- Half of the **cases of acute ITP follow a viral infection**; symptoms appear abruptly about 1 – 2 weeks later (viral illness may trigger development of autoantibodies by uncertain mechanisms)
- **Diagnosis of exclusion**

Case Question

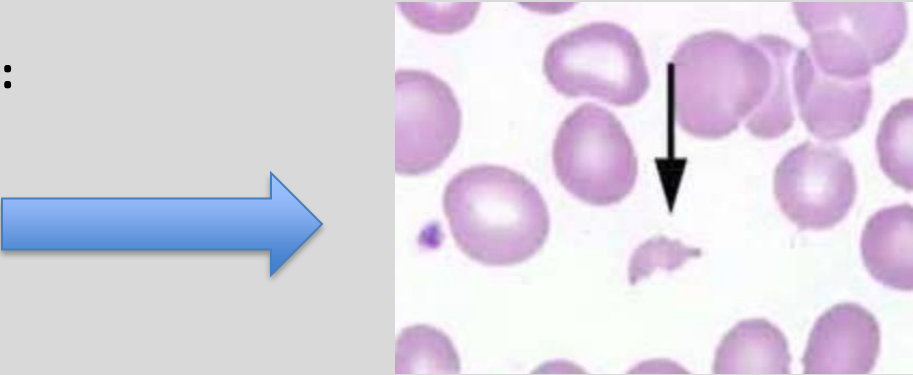
A 9 year old girl develops widespread pinpoint skin hemorrhages. She recovered from a flu-like illness one week earlier. Lab findings show a platelet count of 20,000 but no other abnormalities. A bone marrow biopsy reveals increased number of megakaryocytes. Without treatment, after 2 months, her platelet count is back to normal. What most likely caused her thrombocytopenia?

- A. Decreased production of platelets
- B. Deficient platelet receptors
- C. Autoantibodies against platelets
- D. Viral destruction of platelets
- E. Thrombocytopenia secondary to leukemia

THROMBOTIC MICROANGIOPATHIES:

THROMBOTIC THROMBOCYTOPENIA PURPURA (TTP)

HEMOLYTIC UREMIC SYNDROME (HUS)

- **TTP and HUS:**
 - Excessive activation of platelets which deposit as thrombi in small blood vessels
 - Intravascular thrombi cause microangiopathic hemolytic anemia and widespread organ dysfunction with platelet consumption leading to thrombocytopenia
 - Activation of coagulation cascade **not** primary importance; PT and PTT usually normal
 - **TTP** traditionally defined as **pentad**:
 - Fever
 - Thrombocytopenia
 - Microangiopathic hemolytic anemia
 - Transient neurologic deficits
 - Renal failure
- 
- **HUS** also associated with microangiopathic hemolytic anemia and thrombocytopenia but usually no neurologic symptoms; prominence of acute renal failure; frequent occurrence in children
 - Distinctions blurred; many adult patients with “TTP” lack one or more of the five criteria; some patients with “HUS” have fever and neurologic dysfunction

TTP - Platelet Deficiency (Quantitative) – 1° Hemostasis

- Pathophysiology: Inhibition (acquired antibody) OR deficiency (genetic/inherited) of enzyme **ADAMTS13** : metalloprotease that degrades vWF multimers
- If enzyme is either deficient or inhibited, large vWF multimers accumulate in plasma, promoting spontaneous platelet activation, adhesion, and aggregation resulting in **MICROTHROMBI FORMATION**
- vWF multimers in areas of high shear stress lead to small vessel obstruction
- Labs: **Low platelets/ Normal PT/PTT, Low Hb, Schistocytes “helmet cells” on smear**
- **Plasma exchange removes autoantibodies**, provides functional ADAMTS13; TTP can be treated > 80% of patients (once fatal)

TTP: Thrombotic Thrombocytopenic Purpura – Labs & Treatment

- Traditionally defined as pentad:
 - Fever
 - Microangiopathic hemolytic **Anemia** (**schistocytes** on blood smear, increased **LDH**)
 - Thrombocytopenia (low platelets)
 - Renal failure (**elevated creatine**)
 - Neurological deficits (headache, confusion, seizure)
 - PT/PTT will be **normal**
- Treatment: plasma exchange, steroids, rituximab

HUS - Platelet Deficiency (Quantitative) – 1° Hemostasis

Hemolytic uremic syndrome

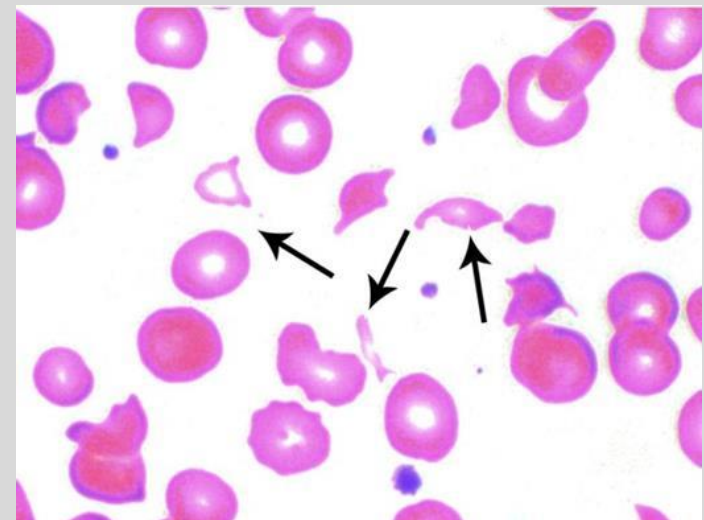
- Normal levels of ADAMTS13:
- **Typical HUS:** strongly associated with **infectious gastroenteritis caused by Escherichia coli strain O157:H7 enterohemorrhagic** (undercooked meat) - produces a **shiga-like toxin** which causes endothelial injury, provoking **platelet activation and aggregation**
 - Children and older adults at highest risk, often present with **bloody diarrhea**; few days later HUS develops
- **Atypical HUS: associated with defects in alternative complement pathway inhibitor deficiencies**
 - **Complement factor H**
 - **membrane cofactor protein (CD46), or factor I** (proteins that prevent excessive activation of alternative complement pathway)
- May be caused by inherited defects or acquired inhibitory autoantibodies; associated with remitting, relapsing clinical course

HUS- Platelet Deficiency (Quantitative) - 1° Hemostasis

- Basis for platelet activation in typical and atypical HUS unclear; therapeutic antibodies that inhibit activation of complement factor C5 effective in preventing thrombosis in atypical HUS
- **Typical HUS treated supportively**; patients who survive acute insult usually recover, some have permanent renal damage, require dialysis or renal transplantation
- Shiga-like toxin vs ADAMTS13 assay for diagnosis
- Labs: **Low platelets, Normal PT/PTT, Low Hb, Schistocytes, renal dysfunction (elevated BUN and Cr)**
- Certain drugs and radiation may also result in thrombotic microangiopathies resembling HUS because of damage to endothelial cells

HUS: hemolytic uremic syndrome– Presentation

- Epidemiology: pediatric population
- Presentation: Triad (3)
 - Renal failure (**elevated creatine**)
 - Microangiopathic hemolytic **Anemia** (**schistocytes** on blood smear, increased **LDH**)
 - Thrombocytopenia (low platelets)
 - Bloody diarrhea
 - PT/PTT will be **normal**
- **Treatment:** supportive care



SUMMARY

Thrombotic Thrombocytopenic Purpura

- Deficiency of ADAMTS13

- Inherited
- Acquired (autoantibodies)

Hemolytic Uremic Syndrome

- *Typical: Escherichia coli strain O157:H7 infection*

- Endothelial damage by Shiga-like toxin

- *Atypical: alternative complement pathway inhibitor deficiencies (complement factor H, membrane cofactor protein [CD46], or factor I)*

- Inherited
- Acquired (autoantibodies)

- Miscellaneous associations

- Drugs (cyclosporine, chemotherapeutic agents)
- Radiation, bone marrow transplantation
- Other infections (HIV, pneumococcal sepsis)
- Conditions associated with autoimmunity (systemic lupus erythematosus, HIV infection, lymphoid neoplasms)

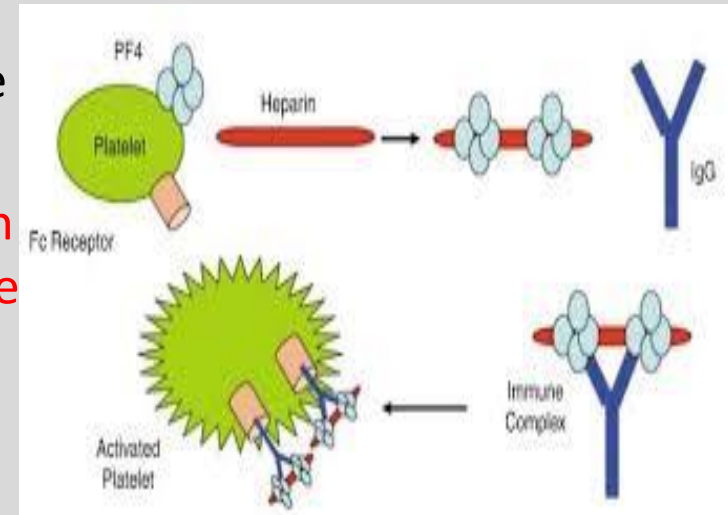
Case Question

A 27 year old previously healthy woman develops spontaneous vaginal bleeding and the following day experience a tonic-clonic seizure. On PE, she is febrile, jaundiced, and has widespread purpura. Lab findings include a hemoglobin of 5.3 g/dl with 8% reticulocytes, a platelet count of 10,000/ul and a BUN of 48 mg/dl. Her peripheral blood smear shows many schistocytes. What is the most likely mechanism for her thrombocytopenia?

- A. Shiga-like toxin from E. coli O157:H7
- B. Autoantibodies against platelet receptors
- C. Defect in complement factor H
- D. Deficiency in plasma enzyme ADAMTS13
- E. Chronic renal failure

Heparin Induced Thrombocytopenia

- Thrombocytopenia in about 5% of persons receiving heparin
- Type 1 typically clinically silent
- Type 2 HIT: less common, often life threatening
 - Occurs 5 to 14 days after therapy begins (some sources 2 – 5 days following heparin dose)
 - IgG antibodies against **COMPLEX of heparin bound to platelet factor 4 (PF4) (membrane glycoprotein)**
 - Binding of antibody to these complexes activates platelets and promotes thrombosis, even though thrombocytopenia present (paradoxical)
 - develop **petechial rash, thrombocytopenia**
- Immediately stop heparin - start an alternative nonheparin anticoagulant



Qualitative Platelet Disorders

Bernard-Soulier

Glanzmann's Thrombasthenia

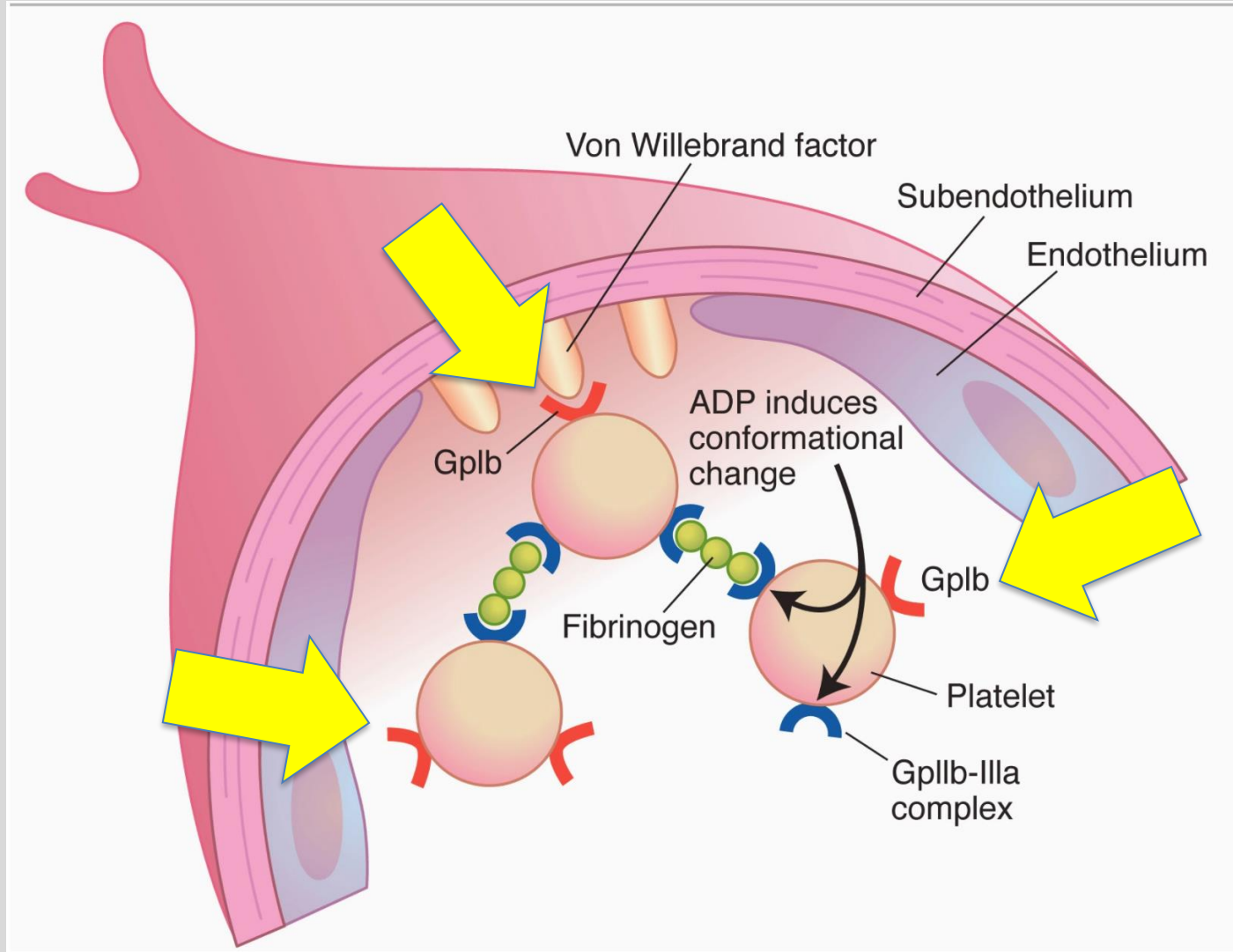
Disorders of PLT secretion

Aspirin effects on PLT function

Uremia and PLT function

Bernard-Soulier: Platelet Dysfunction (Qualitative) - 1° Hemostasis

- Autosomal recessive defect in **PLATELET ADHESION** (to subendothelial matrix/collagen) – no platelet plug
- Caused by inherited deficiency of platelet membrane **glycoprotein complex Ib (GpIb)** (receptor for vWF needed for normal platelet adhesion to subendothelial extracellular matrix)
- **DECREASED** expression of GpIb leads to **DECREASED** adhesion of **PLATELET (GpIb)** to vWf
- Defect in platelet plug formation due to impaired **ADHESION** of platelets to subendothelial collagen (via vWF)

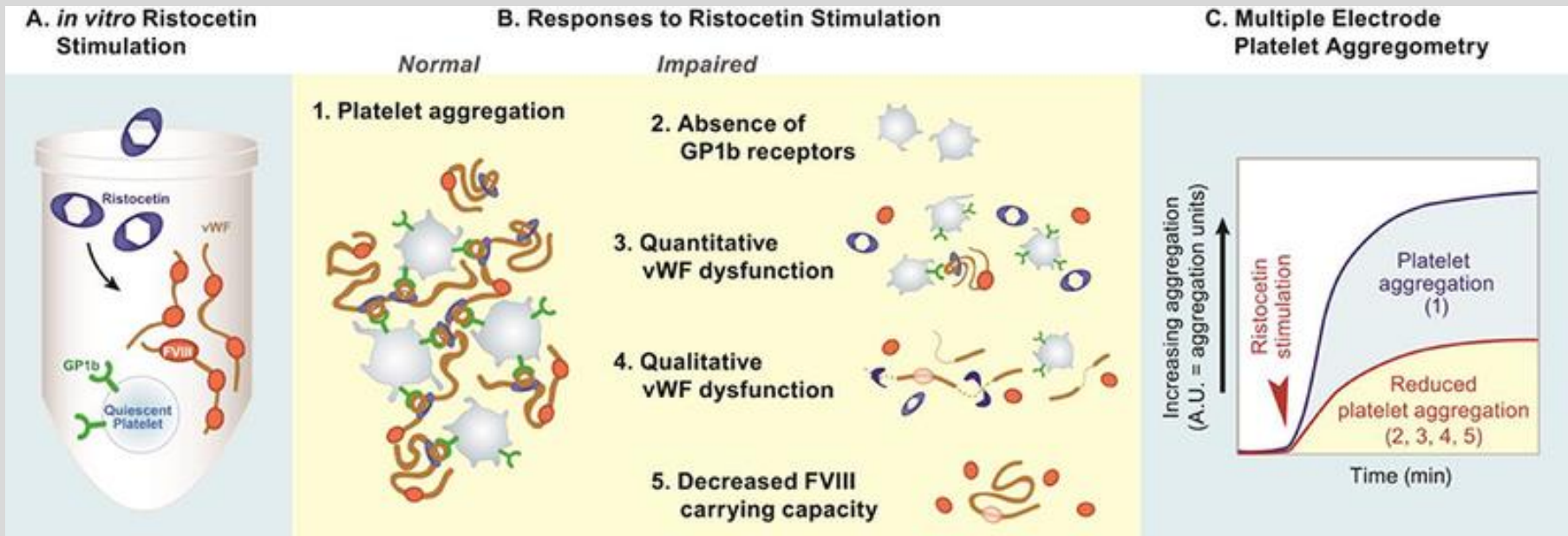


Bernard-Soulier Syndrome – Presentation

- Normal or decreased platelet count
- Large platelets in peripheral blood
- Bleeding time increased
- Affected patients: variable, often severe, bleeding tendency
- **Also:** No **agglutination with ristocetin** (antibiotic, causes von Willebrand factor to bind platelet receptor [GpIb] resulting in agglutination, measured by aggregometry)
 - If there's no GpIb... what can vWF bind to? Nothing
 - *Note:* Flow cytometry of platelet glycoprotein is the confirmatory test. It demonstrates marked reduction of CD42a (GPIX) and CD42b (GPIb-alpha)

<https://www.ncbi.nlm.nih.gov/books/NBK557671/#:~:text=The%20most%20common%20differential%20diagnosis,clinches%20the%20diagnosis%20of%20BSS.>

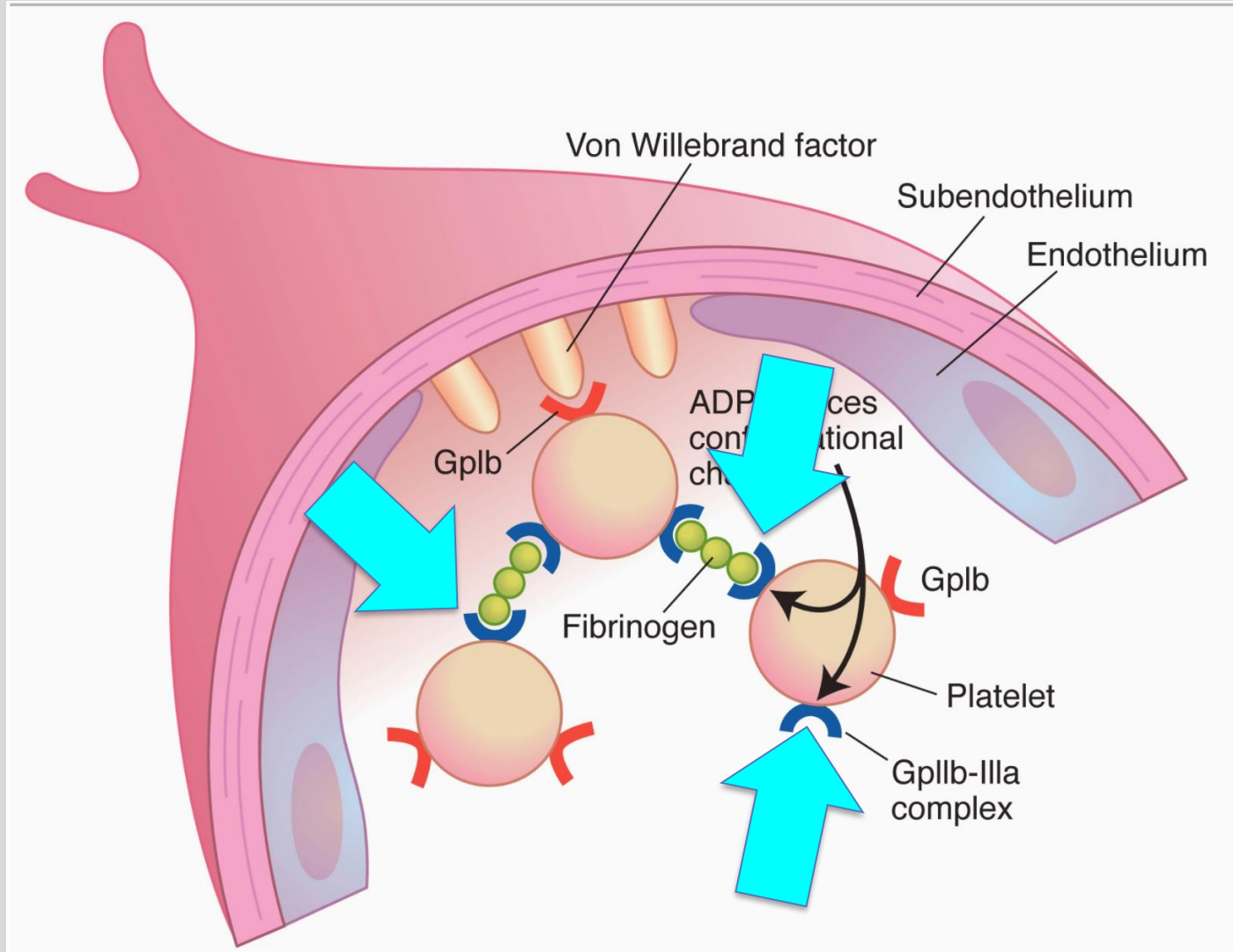
The antibiotic ristocetin causes von Willebrand factor to bind the platelet receptor glycoprotein Ib (GpIb), so when ristocetin is added to normal blood, it causes agglutination.



Glanzmann thrombasthenia: Platelet Dysfunction (Qualitative) - 1° Hemostasis

Glanzmann's Thrombasthenia – defective platelet aggregation

- Autosomal recessive defect in **aggregation**
- **Decreased GpIIb/IIIa expression**
- **Decreased platelet-to-platelet aggregation and defective platelet plug formation**
- **Platelets fail to aggregate** in response to adenosine diphosphate (ADP), collagen, serotonin, epinephrine, or thrombin because of deficiency or dysfunction of **glycoprotein IIb–IIIa** (integrin involved in “bridge formation” between platelets by binding fibrinogen)
- **PLT count usually normal**
- **Increased bleeding time**
- Bleeding often **severe**
- Agglutination with ristocetin*
 - Why? If there is still GpIb!! vWF can bind!



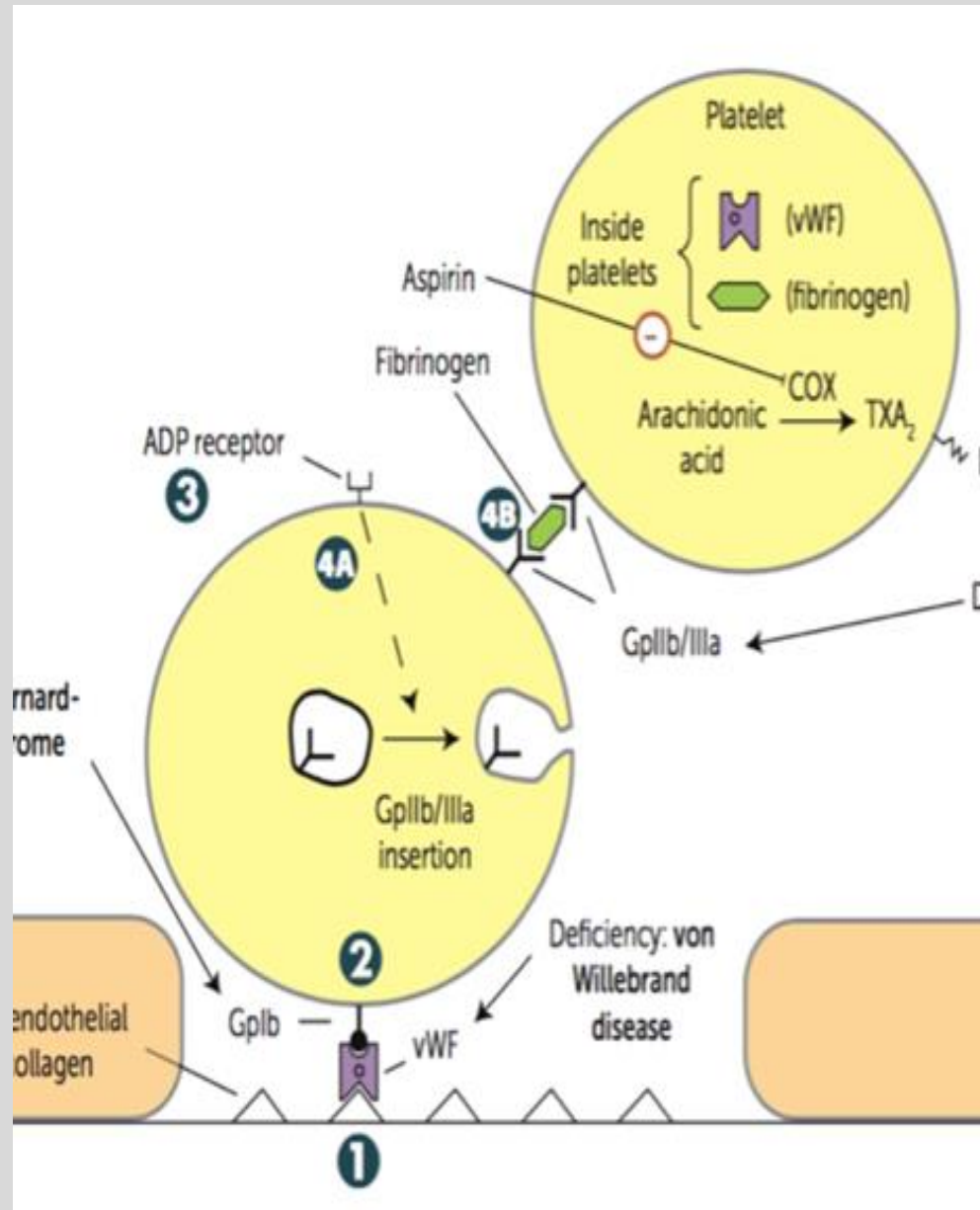
Note: Glanzmann thrombasthenia is definitively diagnosed by tests that determine if there is a deficiency of the $\alpha\text{IIb}\beta 3$ (GPIIb/IIIa) receptor

Disorders of Platelet Secretion and Acquired Defective Platelet Function

- **Storage pool diseases (disorders of platelet secretion):**
Defective release of certain mediators of platelet activation; various, complex biochemical defects
- Ingestion of **aspirin**
- **Uremia:** second common condition associated with acquired defects in platelet function

Aspirin: Thromboxane A₂ TXA₂

- Endothelial cells make TXA₂ via **COX pathway**
- **TXA₂** receptors found on platelets
- TXA₂ binds to receptors on PLTs, **activating them** (shape change)
- **ASA inhibits COX pathway (for lifetime of PLT)** leading to decreased PLT function
- **Anti-platelet effects of aspirin last 7-10 days (life span of platelet)**; basis for use in preventing coronary thrombosis



Uremia

- Renal dysfunction leads to a build up nitrogenous waste products (cannot be excreted)
- Platelet dysfunction caused by **uremic toxins** in plasma
- Poor **aggregation and adhesion** of platelets and interaction with endothelium
- **Prolonged bleeding time**
- Normal PLTs
- Normal coagulation testing

Clotting factor deficiencies/dysfunctions

Mixed-platelet disorders

**Hemophilia A, B and C
von Willebrand's Disease**

**Disseminated Intravascular Coagulopathy
(Consumptive Coagulopathy)**

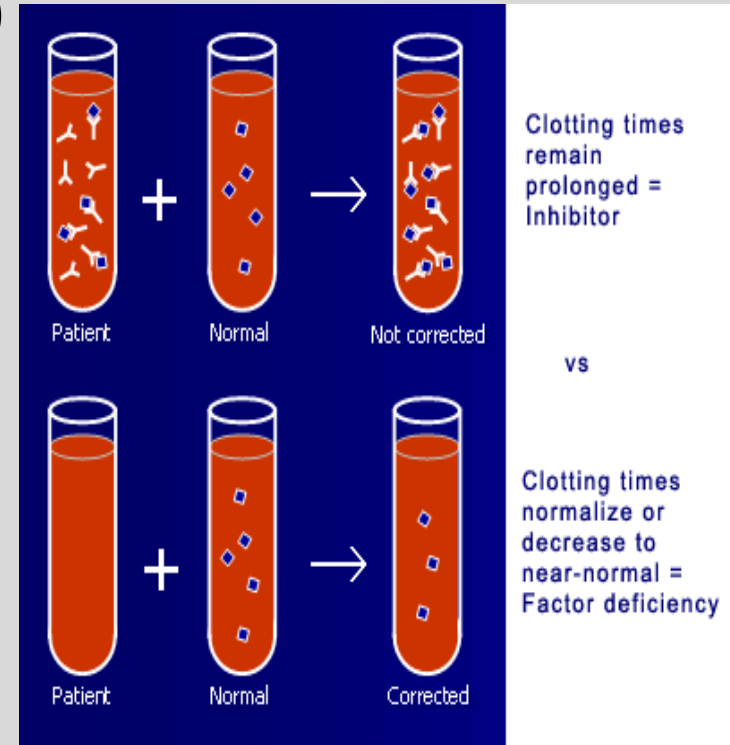
Vitamin K deficiency

Coagulation Disorders: general etiologies and diagnosis

- Coagulation disorders may be due to clotting factor deficiencies or acquired factor inhibitors (most commonly against **factor VIII**)
- Normal Platelet count, but increased PT or PTT or both**
 - Why? Coagulation (clotting) factors of cascade affected; assessed with PT/PTT

Often diagnosed with a **mixing study**:

- Normal plasma** added to patient's plasma. Would allow any **clotting factors deficiencies to correct** (the PT or PTT returns to within normal range)
- Acquired factor inhibitors... would NOT correct**
 - Why? Because autoantibodies or factor inhibitors within patient's plasma; adding more factors just results in them being inhibited as well



Clotting Factor Dysfunction - 2° Hemostasis

General Comments

- Bleeding due to coagulation factor deficiencies:
 - Large posttraumatic ecchymoses or hematomas
 - Prolonged bleeding from laceration or after surgical procedure (e.g., tooth extraction)
- Bleeding often into GI and urinary tracts and into weight-bearing joints (hemarthrosis), unlike thrombocytopenia

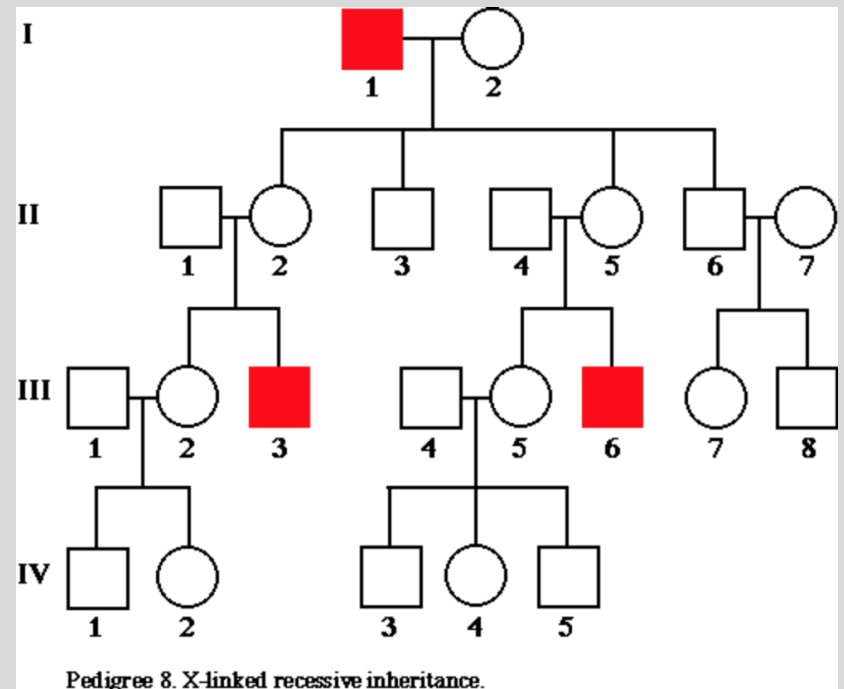
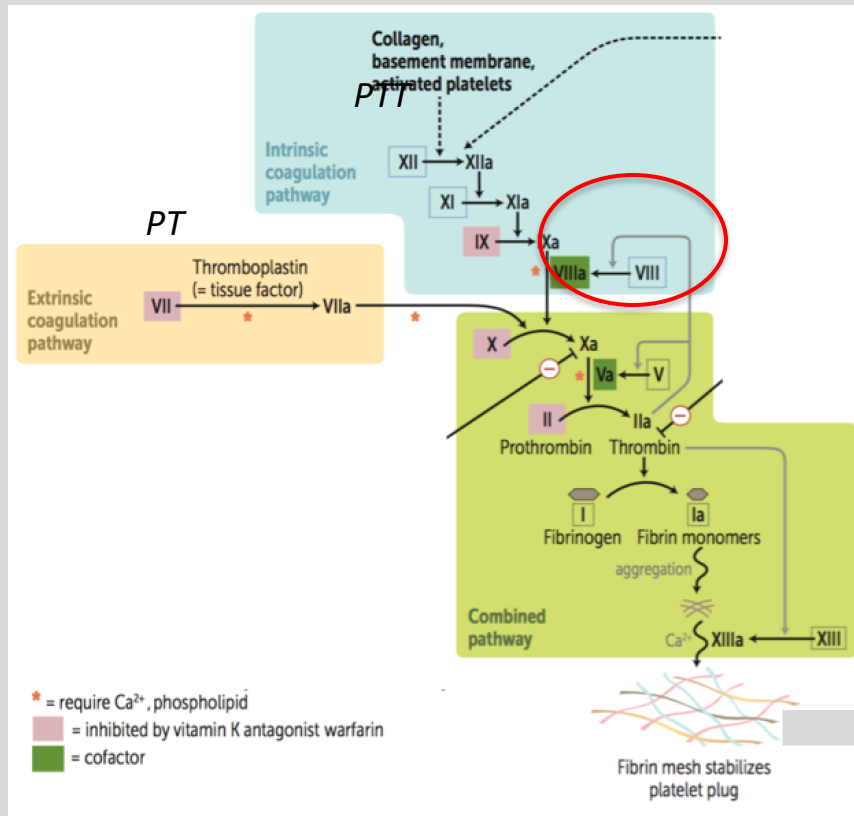


Hemophilia A: Clotting Factor Dysfunction - 2° Hemostasis

- Deficiency of **factor VIII**, X-linked recessive
- Any **intrinsic pathway coagulation defect** will specifically **elevate PTT**
- **Most common hereditary disease associated with life-threatening bleeding; caused by mutations in factor VIII**
- **X-linked recessive trait:** affects mainly males and homozygous females; about 30% of patients have no family history (new mutations)
- Wide range of clinical severity, correlates with level of factor VIII activity:
 - < 1% of normal levels: severe disease
 - Symptomatic cases:
 - Easy bruising and massive hemorrhage after trauma or operative procedures
 - “spontaneous” hemorrhages in body areas susceptible to trauma (esp. joints – hemarthroses)
 - Excessive bleeding after circumcision or umbilical cord separation
 - Petechiae usually not present

Clotting Factor Dysfunction - 2° Hemostasis

- Hemophilia A – Factor VIII – Classic Hemophilia
- X-linked Recessive



30% of cases can result from spontaneous mutation and have no Family History

Hemophilia A: Clotting Factor Dysfunction - 2° Hemostasis

- Main concern hemophilia patient: **hemorrhage**
- **Prolonged PTT and a normal PT** (abnormality of intrinsic coagulation pathway)
- **Hemarthroses:** bleeding into the joints, most common is knee; easy bruising, bleeding after trauma or surgery (dental procedures)
- Factor VIII–specific assays needed for diagnosis
- Traditionally treated with infusions of recombinant factor VIII
- About **15% of patients develop antibodies that bind and inhibit factor VIII**
- **Treatment:** desmopressin, factor VIII concentrate, emicizumab
 - **Desmopressin increases vWF** release which increases functional levels of factor VIIIa
- Before development of recombinant factor VIII therapy, thousands of hemophiliacs received plasma-derived factor VIII concentrates containing HIV, many developed AIDS

Hemophilia B: Clotting Factor Dysfunction - 2° Hemostasis

- Aka **Christmas Disease** (named after first patient identified, not holiday):
- Deficiency of **factor IX**, **X-linked recessive**, clinical severity varies
- **Prolonged PTT** and a **normal PT** (abnormality of intrinsic coagulation pathway)
- **Diagnosis**: assay of IX factor levels
- **Hemarthroses**: bleeding into the joints, most common is knee; easy bruising, bleeding after trauma or surgery (dental procedures)
- **Severe factor IX deficiency - clinically indistinguishable from factor VIII deficiency (hemophilia A)** (factors VIII and IX function together to activate factor X)
- Many mutations involving gene that encodes factor
- In about 15%, factor IX protein present but not functional
- Treated with infusions of recombinant factor IX
- Before development of recombinant factor VIII therapy, thousands of hemophiliacs received plasma-derived factor VIII concentrates containing HIV, many developed AIDS

Hemophilia C: Clotting Factor Dysfunction

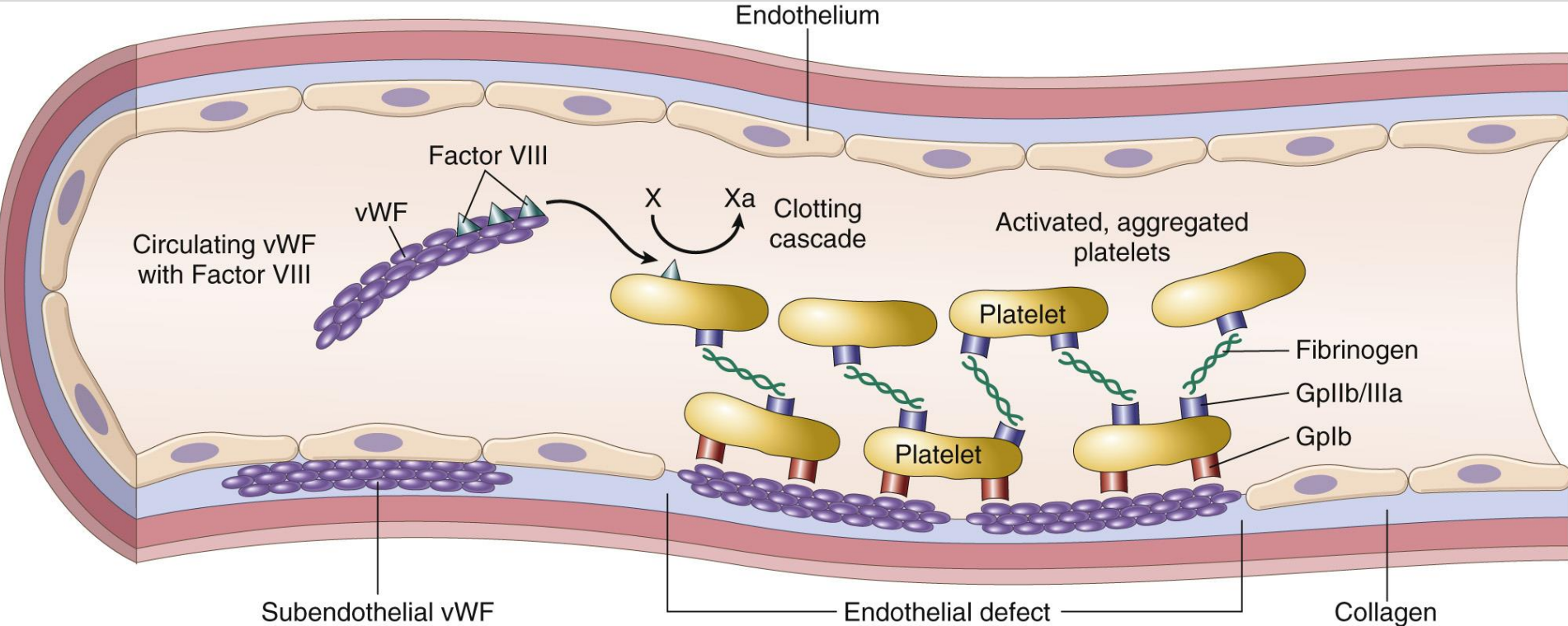
- Any **intrinsic pathway coagulation defect** will specifically **elevate PTT**
- Deficiency of **factor XI**, **AUTOSOMAL recessive** – **more mild disease**
- Main concern: **hemorrhage**
- **Prolonged PTT and a normal PT** (abnormality of intrinsic coagulation pathway)
- **Hemarthroses**: bleeding into the joints, most common is knee; easy bruising, bleeding after trauma or surgery (dental procedures)
- **Treatment**: infusions of recombinant factor XI

Von Willebrand Factor - Review

- Most important hemostatic function of vWF : **promotes adhesion of platelets to subendothelial matrix via bridging interactions between platelet glycoprotein Ib–IX, vWF, and matrix components (e.g. collagen)**
- **Factor VIII binds to and is stabilized by vWF :** increases half-life of factor VIII from 2.4 hours to about 12 hours
- vWF secreted as multimers into circulation by endothelial cells

Factor VIII and vWF

- Factor VIII function measured by coagulation assays with mixtures of patient plasma and factor VIII–deficient plasma
- vWF function assessed using ristocetin agglutination test
 - add patient plasma to normal formalin-fixed platelets with ristocetin
 - small molecule binds and “activates” vWF
 - ristocetin induces multivalent vWF multimers to bind platelet glycoprotein Ib–IX, form interplatelet “bridges”: results in clumping (agglutination) of platelets
- von Willebrand disease patients have defects in platelet function but **normal platelet counts**
- **Plasma level of active vWF, measured as ristocetin cofactor activity, is reduced (no agglutination)**



- Factor VIII made in liver (endothelial cells) and kidney, essential cofactor of factor IX
- vWF made in endothelial cells and megakaryocytes; form complex in blood; also present in subendothelial matrix of normal blood vessels and α -granules of platelets
- **Following endothelial injury: exposure of subendothelial vWF causes adhesion of platelets, mostly via glycoprotein Ib (GpIb) platelet receptor;** circulating vWF and vWF released from α -granules of activated platelets bind exposed subendothelial matrix, further contributing to platelet adhesion and activation
- Activated platelets form hemostatic aggregates; fibrinogen participates in aggregation via bridging interactions with glycoprotein IIb/IIIa (GpIIb/IIIa) platelet receptor

Von Willebrand Disease: Clotting Factor Dysfunction - 2° Hemostasis

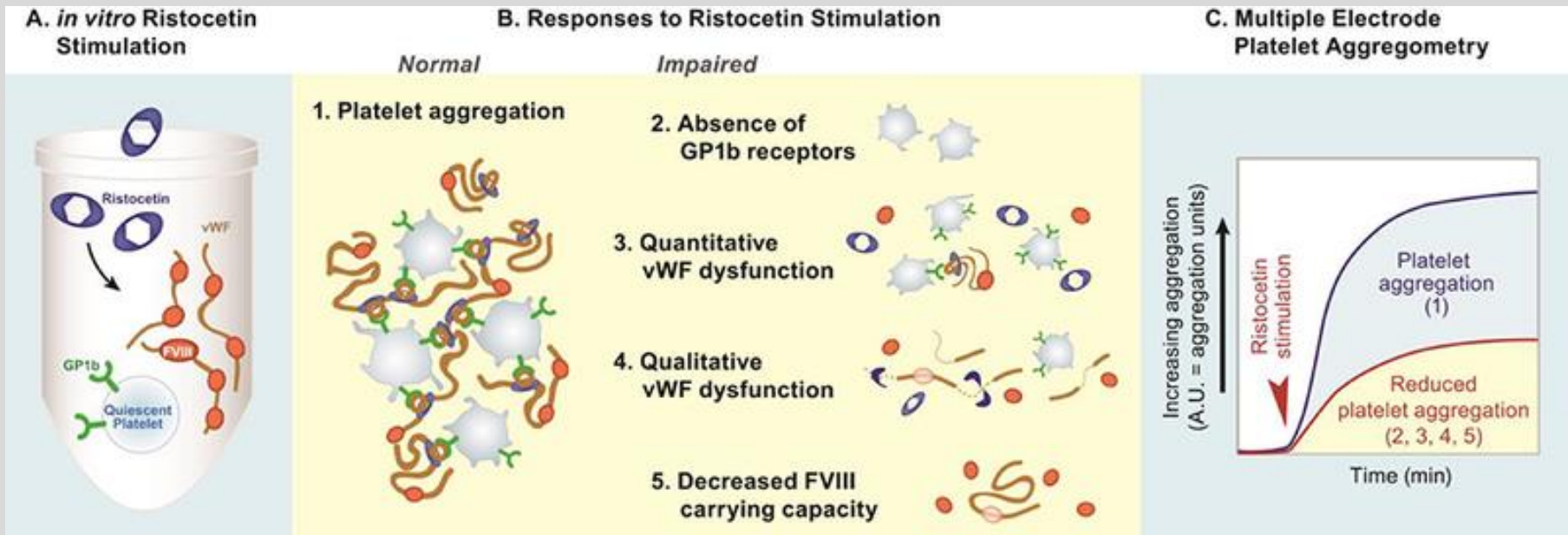
- **Most common inherited bleeding disorder of humans, about 1% of adults in US**
- Bleeding usually mild, may go unnoticed until surgery or dental procedure
- Spontaneous bleeding from mucous membranes (e.g., epistaxis), excessive bleeding from wounds, or menorrhagia
- Usually autosomal dominant, but rare autosomal recessive variants exist
- Hundreds vWF variant described, only few cause disease; 3 types recognized:
- **Type 1 and type 3 von Willebrand disease associated with **QUANTITATIVE** defects in vWF**
 - **Type 1, autosomal dominant** (incomplete penetrance; variable expressivity): mild to moderate vWF deficiency (70% of case); associated with many mutations
 - **Type 3, autosomal recessive: little to no vWF synthesis**; factor VIII levels also reduced (vWF stabilizes factor VIII in circulation); **bleeding often severe**

Von Willebrand Disease: Clotting Factor Dysfunction – Presentation

- Type 1 and type 3 von Willebrand disease: **prolonged PTT (deficiency of vWF decreases stability of factor VIII)**
- Wide variability in clinical expression common
- **Easy bruising** (occurs with very little or no trauma/injury)
- **Bleeding time: INCREASED**
- **Normal platelet count (usually)**
- **Ristocetin cofactor assay: no platelet aggregation**

Ristocetin (antibiotic) **causes von Willebrand factor to bind platelet receptor glycoprotein Ib (GpIb)**, so when **ristocetin is added to normal blood, it causes agglutination**

vWF disease: NO platelet aggregation, because vWF not present in sufficient quantities!



Von Willebrand Disease: Clotting Factor Dysfunction – Pathogenesis (Qualitative)

- Type 2 von Willebrand disease:
 - **QUALITATIVE** defects in vWF; several subtypes
 - **type 2A most common; autosomal dominant**
 - vWF expressed in normal amounts, but missense mutations lead to **defective multimer assembly resulting in absence of large- and intermediate-sized multimers** (most active forms of vWF)
 - 25% of all cases
 - mild to moderate bleeding

Disseminated Intravascular Coagulation

- Acute, subacute, or chronic thrombohemorrhagic disorder with **excessive activation of coagulation and formation of thrombi in microvasculature**
- **Not a primary disease; acquired coagulopathy from different conditions**
- Secondary complication of many disorders: most commonly associated with **obstetric complications, malignant neoplasms, sepsis, and major trauma**
- Consumption of platelets, fibrin, and coagulation factors and, secondarily, activation of fibrinolysis
- Can present with signs and symptoms relating to tissue hypoxia and infarction caused by microthrombi; hemorrhage, due to depletion of factors and activation of fibrinolytic mechanisms; or both

Disseminated intravascular coagulation

- Two major mechanisms trigger DIC:
 - (1) release of tissue factor or other procoagulants into circulation, and
 - (2) widespread injury of endothelial cells

CAUSATIVE AGENTS (examples):

- **Procoagulants** (e.g., tissue factor) from, e.g.: placenta in obstetric complications; tissues injured by trauma or burns; mucus/mucin released from certain adenocarcinomas (may directly activate factor X); **APL (acute promyelocytic leukemia)**
- **Endothelial injury** can initiate DIC; injuries that cause endothelial cell necrosis expose subendothelial matrix - leads to activation of platelets and coagulation pathway
- **TNF - one mediator of endothelial injury (implicated in DIC occurring with sepsis)**; induces endothelial cells to express tissue factor and decreases expression of thrombomodulin - tilts balance towards coagulation
- **Widespread endothelial injury** can also occur with deposition of antigen-antibody complexes (e.g., systemic lupus erythematosus), temperature extremes (e.g., heat stroke, burns), or microorganisms (e.g., meningococci, rickettsiae)

DIC

- Consequences of DIC:
 - **Widespread deposition of fibrin thrombi within microcirculation:** leads to ischemia of vulnerable organs and **microangiopathic hemolytic anemia** (from fragmentation of rbc's as they squeeze through narrowed microvasculature)
 - **Consumption of platelets and clotting factors and activation of plasminogen; leads to bleeding**
- Thrombi: in brain, heart, lungs, kidneys, adrenals, spleen, and liver (decreasing order of frequency) and other organs
- Kidneys: may have small thrombi in glomeruli; severe cases, microinfarcts or bilateral renal cortical necrosis
- Many fibrin thrombi may be in alveolar capillaries, sometimes associated with pulmonary edema and fibrin exudation
- CNS: fibrin thrombi may cause microinfarcts
- **In meningococcemia, fibrin thrombi within microcirculation of adrenal cortex, massive adrenal hemorrhages (Waterhouse-Friderichsen syndrome)**

DIC

- DIC onset: acute/fulminant (e.g., sepsis or amniotic fluid embolism) OR insidious/chronic (e.g., carcinomatosis, retention of dead fetus)
- **Acute DIC, usually associated with obstetric complications or major trauma, dominated by bleeding**
- **Chronic DIC (e.g., cancer patients), usually more thrombosis**
- **Manifestations:** microangiopathic hemolytic anemia; dyspnea, cyanosis, respiratory failure; convulsions and coma; oliguria and acute renal failure; and sudden or progressive circulatory failure and shock
- Prognosis varies, depends on underlying disorder
- Only definitive treatment: remove or treat inciting cause

Disseminated Intravascular Coagulation – Presentation

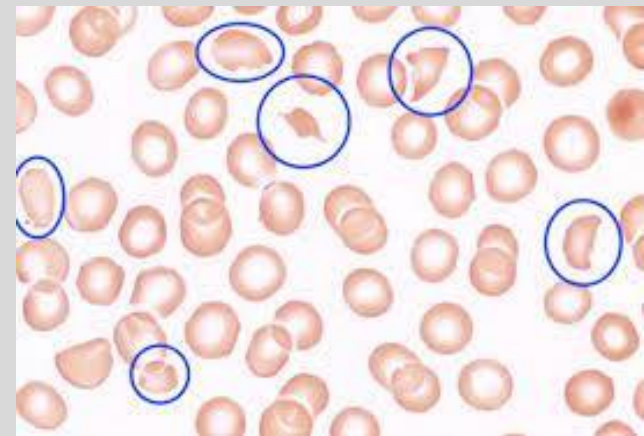
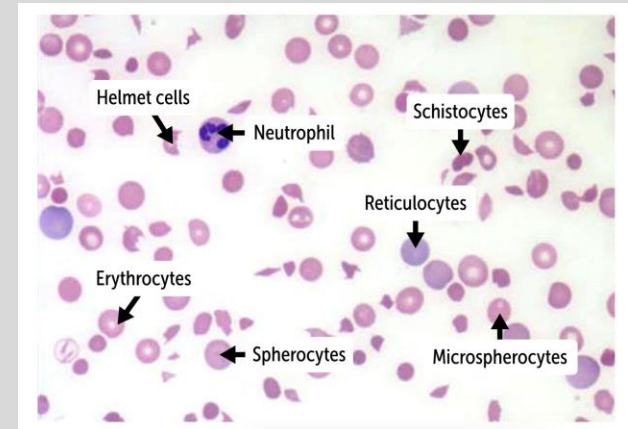
LABS:

- Low platelet count
- Prolonged PT / PTT

– Schistocytes on blood smear (microangiopathic anemia):

Why? RBCs damaged by fibrin thrombi

- Increased fibrin degradation products (D-dimer): **Increased D-Dimers**
- Decreased fibrinogen (note: (Can be elevated – Acute phase reactant))
- Decreased factors 5 and 8 (V and VIII)



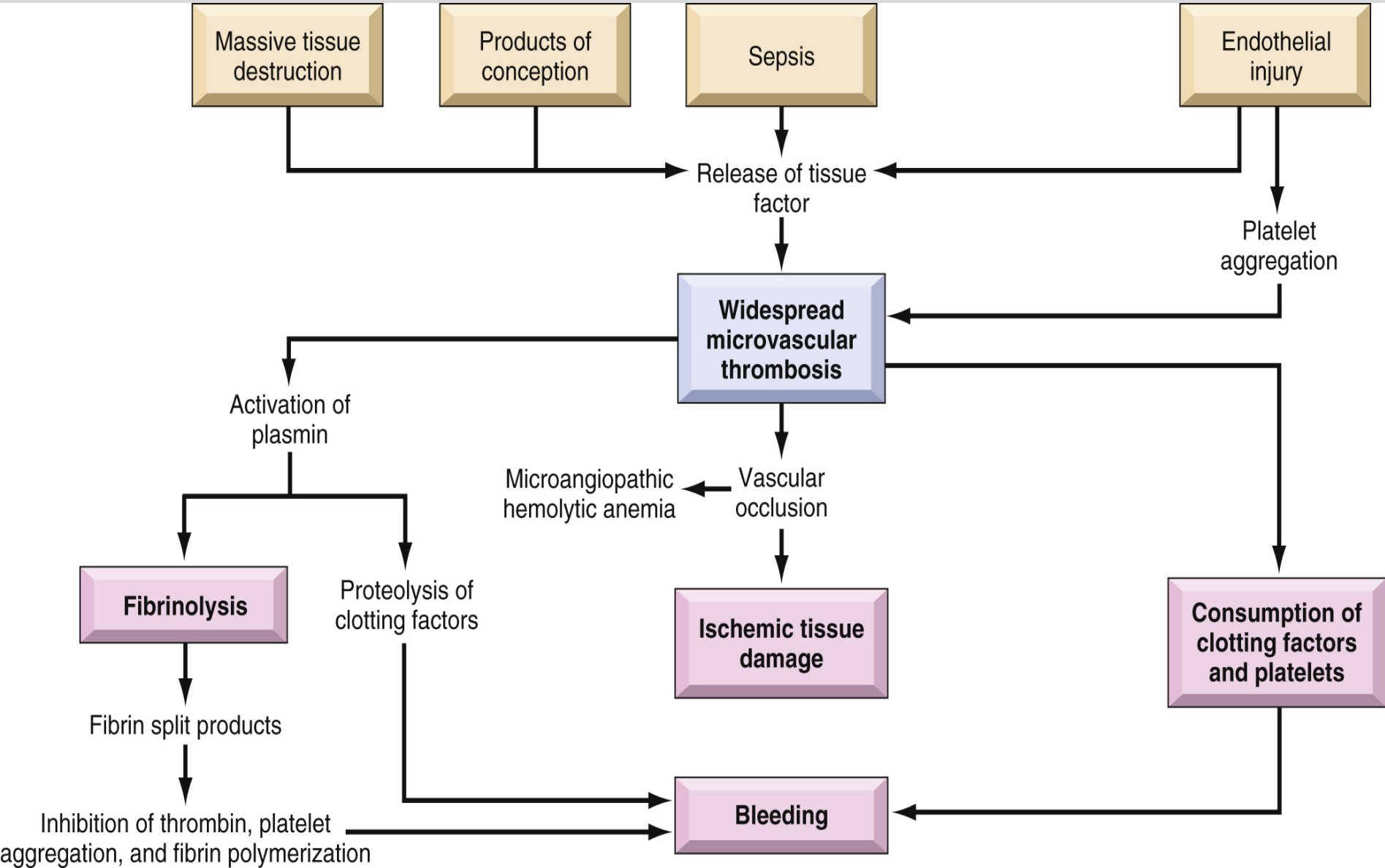


Figure 14.27 Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 14

Pathophysiology of disseminated intravascular coagulation.

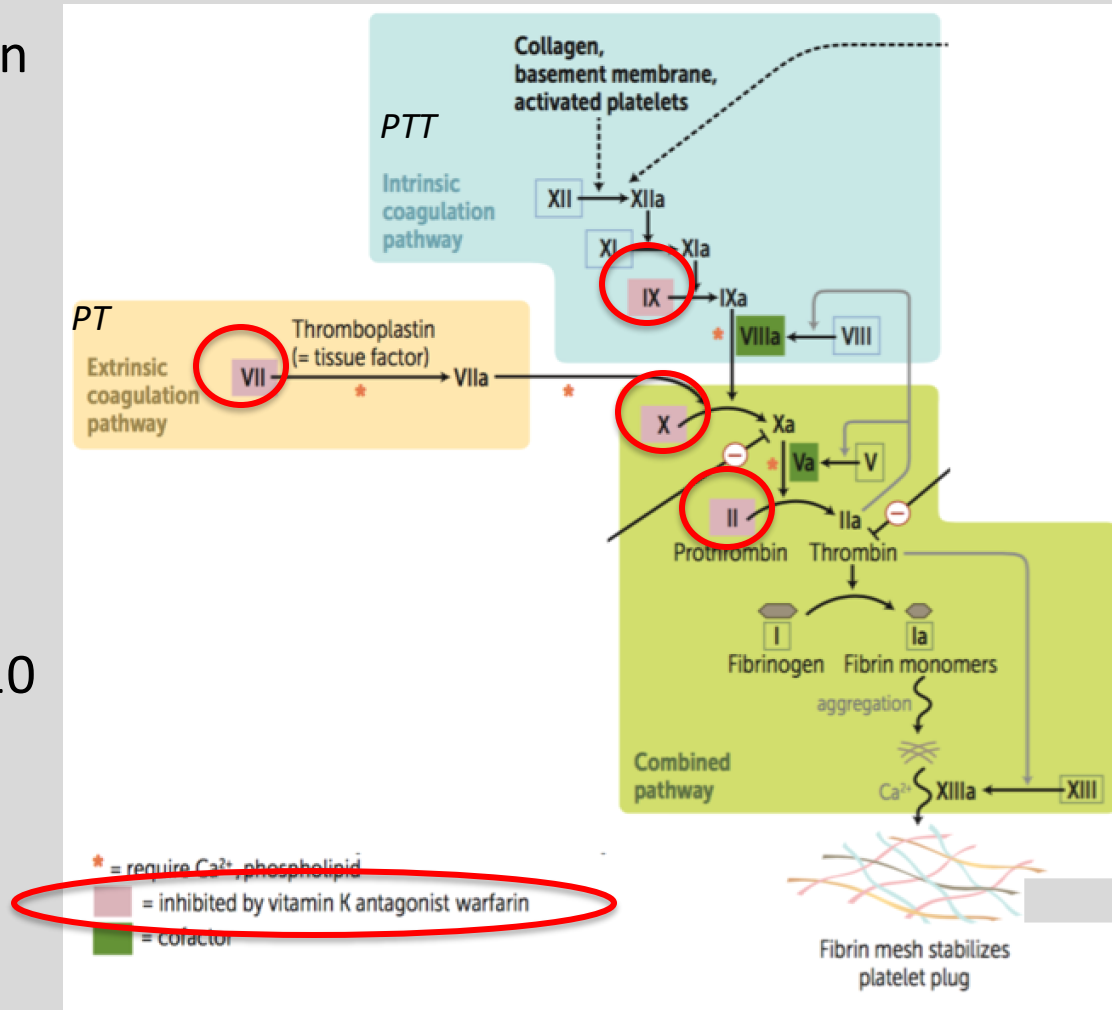
Disseminated Intravascular Coagulation - Pathogenesis

- Supportive care
- **Treat underlying etiology**
- Fresh Frozen Plasmas to replace clotting factors
- RBC transfusion, PLT transfusion (transfusion causes a dilution effect)
- Cryoprecipitate (to replete fibrinogen)

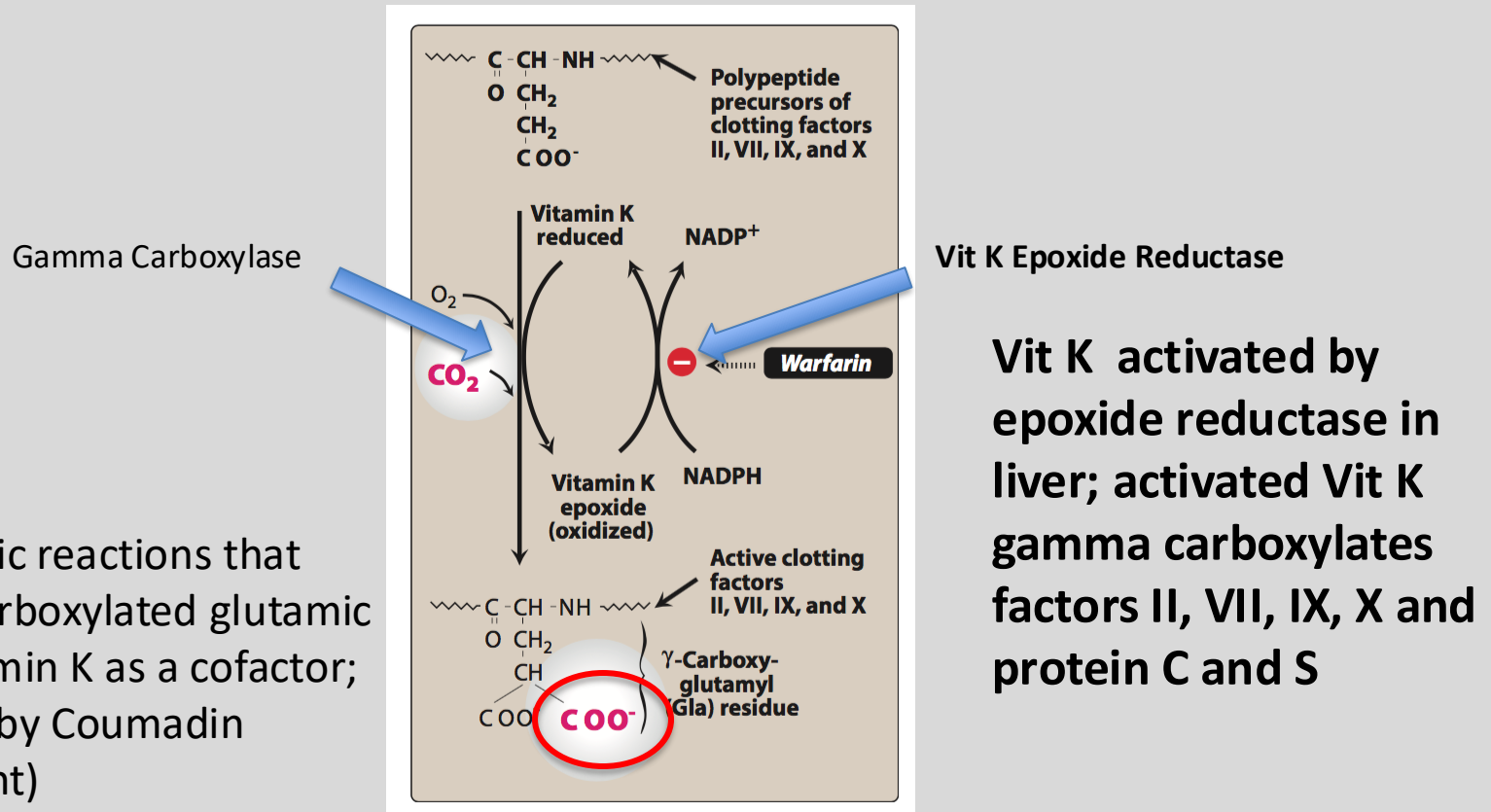
Clotting Factor Dysfunction - 2° Hemostasis

Vitamin K Deficiency – Factors 2, 7, 9, 10

- Vitamin K: fat soluble vitamin made from intestinal flora and found in cabbage, kale, spinach, and egg yolk
- Serves as coenzyme to epoxide reductase in liver
- Helps with carboxylation of glutamic acid residues of clotting factors 2, 7, 9, and 10



- Vitamin K – Factors 2, 7, 9, 10, Pro C, Pro S



Vitamin K Deficiency – Factors 2, 7, 9, 10

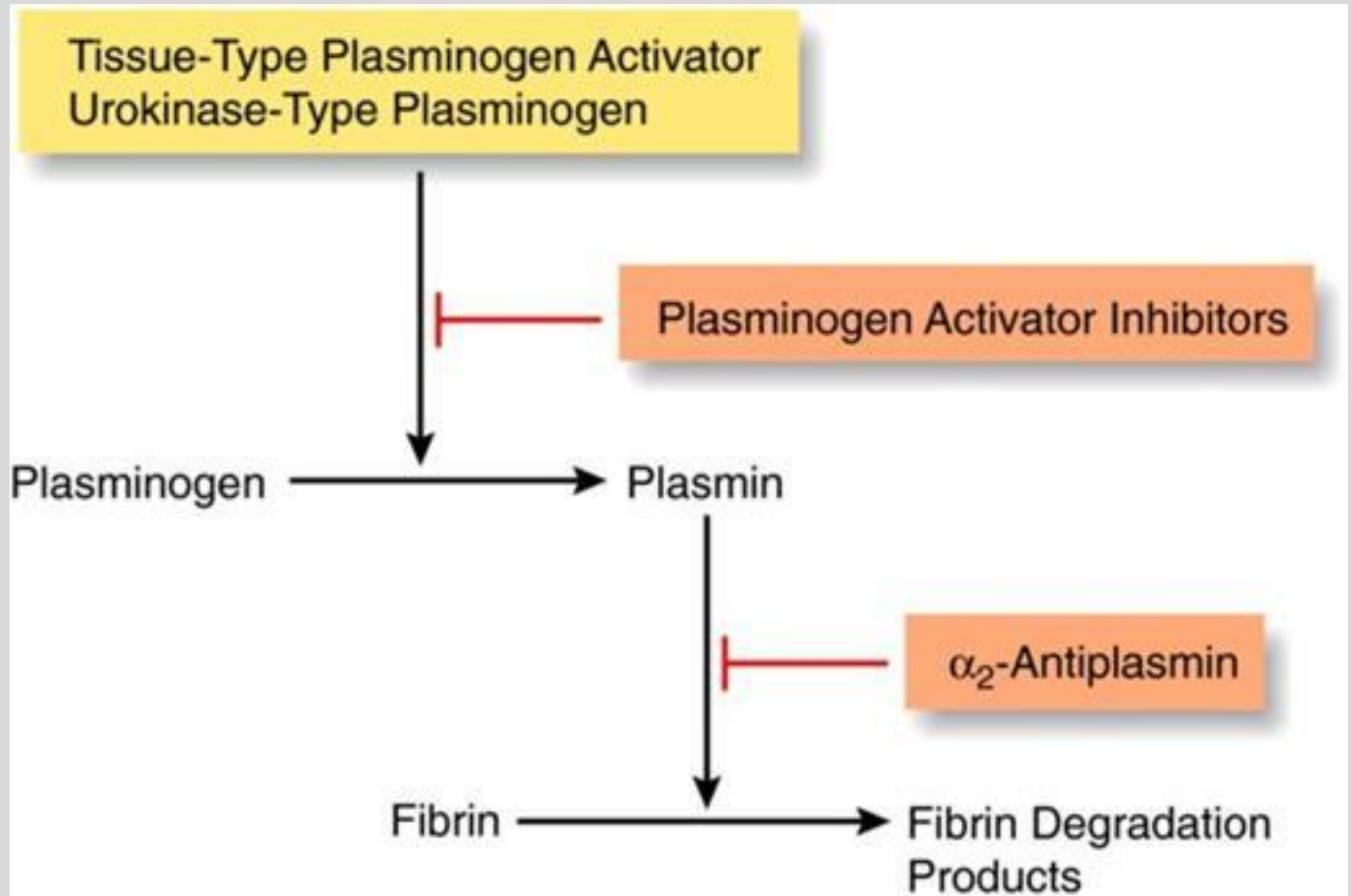
- Similar presentation to other problems of secondary hemostasis
- **Normal Platelet Count & Bleeding Time**
- **Prolonged PT & PTT** (Both intrinsic and extrinsic pathways)
- **Newborns** – (Hemorrhagic Disease of the newborn) due to lack of GI colonization by bacteria that normally synthesize Vitamin K; not found in breast milk; prophylactically inject all newborns at birth with Vitamin K
- **Antibiotic Therapy** – Disrupts vitamin K producing Bacteria
- **Malabsorption Disorders**– Leads to deficiencies in Fat Soluble vitamins A, D, E and K (CF, pancreatic insufficiency, malnutrition, malabsorption syndromes)

Bleeding Disorders: Hemorrhagic Diatheses

Overview of Causes

- Platelet Deficiency (Quantitative)
 - Thrombocytopenia
 - Immune thrombocytopenia purpura (ITP)
 - thrombotic thrombocytopenic purpura (TTP)
 - Hemolytic Uremic Syndrome
- Platelet Dysfunction (Qualitative)
 - Bernard Soulier Disease
 - Glanzmann's Thrombasthenia
 - Disorders of Platelet secretion
 - Aspirin
 - Uremia
- Clotting Factor Deficiency/Dysfunction
 - Von Willenbrand's Disease
 - Hemophilia A / B / C
 - Vitamin K deficiency
 - Disseminated Intravascular Coagulation
- Disorders of Fibrinolysis
 - **Plasmin over activity**
 - **Alpha2-antiplasmin**
- Vessel wall abnormalities
 - Infectious
 - Drug reaction
 - Scurvy / Ehlers-Danlos / Cushings
 - Henoch-Schonlein Purpura
 - Hereditary Hemorrhagic Telangiectasia
 - Perivascular Amyloidosis

Fibrinolysis



Disorders of Fibrinolysis

- Due to plasmin overactivity: resulting in excessive cleavage of serum fibrinogen; examples:
 - Radical prostatectomy – release of urokinase activates plasmin
 - Cirrhosis of liver – reduced production of alpha-2-antiplasmin
- Increased bleeding
- Increase PT/PTT (plasmin destroys coagulation factors)
- Increased bleeding time with normal platelet count (plasmin blocks platelet aggregation)
- Increased fibrinogen split products (d-dimers not formed because fibrin thrombi are absent)
- Treatment: aminocaproic acid which blocks activation of plasminogen

Bleeding Disorders: Hemorrhagic Diatheses

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 - Alpha2-antiplasmin
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 - **Infectious**
 - **Drug reaction**
 - **Scurvy / Ehlers-Danlos / Cushings**
 - **Henoch-Schonlein Purpura**
 - **Hereditary Hemorrhagic Telangiectasia**
 - **Perivascular Amyloidosis**

Bleeding Disorders Caused by Vessel Wall Abnormalities

- **Platelet count, bleeding time, and PT/PTT usually normal**
- Usually not serious bleeding problems; small hemorrhages (petechiae and purpura) in skin or mucous membranes
- Sometimes, more significant hemorrhages into joints, muscles, and subperiosteal locations; or menorrhagia, nosebleeds, gastrointestinal bleeding, or hematuria

Vessel Wall Abnormalities causing bleeding

- **Infections**: Examples: **meningococcemia, other forms of septicemia, infective endocarditis, several rickettsioses**; WHY? - **microbial damage to microvasculature (vasculitis) and disseminated intravascular coagulation**
- **Drug reactions**: cutaneous petechiae and purpura; **vascular injury usually mediated by deposition of drug-induced immune complexes in vessel walls**, leading to hypersensitivity vasculitis
- **Scurvy/Ehlers-Danlos/Cushing syndrome** : microvascular bleeding due to **collagen defects that weaken vessel walls**
- **Henoch-Schönlein purpura**: systemic immune disorder - purpura, colicky abdominal pain, polyarthralgia, acute glomerulonephritis from deposition of circulating immune complexes within vessels throughout body
- **Hereditary hemorrhagic telangiectasia (Weber-Osler-Rendu syndrome)**: autosomal dominant, gene mutations; **characterized by dilated, tortuous blood vessels with thin walls that bleed easily**
- **Perivascular amyloidosis**: misfolded proteins causes damage and vascular fragility, causing bleeding - most common with amyloid light-chain (AL) amyloidosis (mucocutaneous petechiae)

Serious bleeding most often associated with hereditary hemorrhagic telangiectasia

Summary Slide

Review of Hemostasis

- Arteriolar vasoconstriction
- Primary hemostasis – platelet plug
- Secondary hemostasis – coagulation cascade
- Clot stabilization and anti-thrombotic events

Disorders of Primary Hemostasis

Platelet deficiency – quantitative

- Thrombocytopenia

- ITP – acute and chronic, IgG against GpIIb/IIIa

- TTP – microangiopathic anemia, neuro, renal, def. ADAMTS13

- HUS – renal esp., E. coli 0157H7 (typical), defects of complement factor H, etc. (atypical)

Platelet dysfunction – qualitative

- Bernard-Soulier disease – AR, def. GpIb, def. platelet adhesion

- Glanzmann thrombasthenia – AR, def. GpIIb/IIIa def. platelet aggregation

- Disorders of platelet secretion

- Aspirin – acquired, irreversible inhibition of COX

- Uremia

Summary Slide

- Clotting factor deficiencies
 - Von Willebrand disease – vWF bridges collagen to platelets via Gplb, also binds to factor VIII and stabilizes it; PT normal, PTT prolonged
 - Type 1 (AD) and Type 3 (AR) quantitative
 - Type 2 (AD) qualitative
 - Hemophilia A – X-linked recessive, Factor VIII def. , PT normal, PTT prolonged
 - Hemophilia B – X-linked recessive, Factor IX def., PT normal, PTT prolonged
 - Vit. K def. – Factors II, VII, IX, and X, PT and PTT both prolonged
 - Newborns, antibiotics, malabsorption
- DIC – activation of coag. factors, formation of microthrombi, platelets get used up, decreased platelets cause bleeding, thrombotic disorder can become bleeding disorder
 - Not a primary disease, multiple causes
 - Prolonged PT and Prolonged PTT, increased fibrin split products.

Lecture Feedback Form

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

Resources:

- Robbin's 10th edition: chapter 4 and 14
- Rubin's 10th edition: chapter 7, 16, and
- First aid for USMLE Step 1, 2023
- USMLE Rx Bricks

Special thanks to Academic Scholar Samantha Cornwell